

JACOB ENNIS

810-429-7045 • ennis.jacob.pi@gmail.com • linkedin.com/in/jacob-ennis5 • GitHub: gandalfthegreenery • La Mesa, CA

SUMMARY

Machine Learning Engineer with a Master's in Computer Science and experience developing AI-driven software systems for computer vision, data analytics, and autonomous applications. Skilled in Python, machine learning pipelines, cloud-based deployment, and transforming complex data into actionable insights through visualization and decision-support tools. Experienced collaborating across technical disciplines to deliver reliable solutions in operational environments.

Programming: Python, C/C++, SQL, Bash

Machine Learning: PyTorch, TensorFlow, OpenCV, Pandas, Numpy, Scikit-Learn, Object Detection, Semantic Segmentation

Software & Cloud: Git, Docker, Linux/Unix, AWS

Technical Areas: ML pipelines, Data Analysis, Real-Time Systems, Optimization, Sensor Fusion, System Integration

Certificates: OpenCV Master Program - OpenCV, AI/Machine Learning - Arizona State University

PROJECTS

Wearable Tactile Depth Display for Visual Impairments

9/2024-Present

- Developed an AI-enabled perception system that transformed stereoscopic camera data into real-time tactile navigation guidance for assistive mobility applications.
- Built CV and data processing pipelines integrating depth estimation, object detection, and spatial reasoning components.
- Optimized latency, throughput, and inference performance to support responsive operation in dynamic real-world environments.
- Designed and validated end-to-end data flows across perception, decision-making, and hardware feedback systems to improve reliability and usability.

Computer Vision Applications with OpenCV (Classification, Segmentation, Detection)

9/2025-Present

- Developed end-to-end machine learning pipelines for image classification, semantic segmentation, and object detection using PyTorch and OpenCV.
- Built data ingestion, preprocessing, augmentation, and evaluation workflows supporting model development across multiple datasets.
- Achieved approximately 90% mAP for object detection and 82% IoU for semantic segmentation through iterative model optimization.
- Analyzed performance tradeoffs between model accuracy, computational efficiency, and inference latency to guide deployment decisions.

EDUCATION

Master of Science in Computer Science: Arizona State University, Tempe, AZ

5/2022-05/2024

(Statistical Machine Learning, Deep Learning in Visual Computing, Artificial Intelligence)

3.97 GPA

B.S. Physics, Minor in Computer Science: University of Michigan, Ann Arbor, MI

9/2015-12/2019

(Data Structures & Algorithms, Database Management Systems)

3.25 GPA

WORK EXPERIENCE

Blue Guardian, London, ON: Machine Learning Engineer

10/2023-9/2024

- Designed and implemented machine learning pipelines for cognitive distortion detection using deep learning architectures including transformers, graph neural networks, and variational autoencoders.
- Developed Python-based data processing, feature engineering, and model evaluation workflows for noisy real-world datasets.
- Improved model F1 score by 15% and orchestrated a 70% reduction in synthetic/real data drift through systematic experimentation, validation, and performance analysis.
- Collaborated with cross-functional stakeholders to translate research objectives into deployable AI solutions

Locally Grown Salads, North York, ON: Student Software Developer

6/2023-12/2023

- Developed and deployed an AWS-connected edge AI control system integrating real-time sensing, analytics, and automated decision-making for agricultural operations.
- Designed software interfaces between sensors, control systems, and cloud infrastructure to support reliable operation
- Collaborated with multidisciplinary teams to iteratively refine system functionality based on field testing and user feedback.

Monnier Research Lab, University of Michigan, Ann Arbor, MI: Astronomy Research Assistant

10/2015 - 07/2022

- Designed and fabricated precision subsystems and experimental components for infrared astronomical imaging instruments
- Integrated optics, embedded components and closed-loop feedback mechanisms for high-precision experimental platforms
- Supported experimental validation and troubleshooting of complex hardware/software systems under research constraints
- Co-authored peer-reviewed publications related to instrument development and observational data analysis

COMMUNITY ENGAGEMENT

- 2nd Place, Trans-Atlantic AI Hackathon- Developed a CV app for medical device anomaly detection using stereoscopic cameras
- Active member of, and presenter at Google Cloud Developer's Meeting, and San Diego Machine Learning Groups